

# CI/CD Deployment Pipelines and Release Protocols

CLASSIFICATION / ACCESS LEVEL: ENGINEERING & DEVOPS

To maintain software quality and rapid delivery velocity, engineering teams follow strict continuous integration and deployment standards.

**Git Branching Strategy:** We utilize a Trunk-Based Development workflow. Feature branches must be short-lived (less than 48 hours) and require automated linting, unit testing (minimum 80% code coverage), and security vulnerability scanning.

**Pull Request Approvals:** Every Pull Request requires mandatory review and approval from at least two senior engineers before merging into the main branch.

**Deployment Workflow & Staging:** Automated GitHub Actions build container images pushed to AWS ECR. Successful builds deploy automatically to the Dev and Staging clusters. Staging runs automated end-to-end integration tests nightly.

**Production Releases & Rollbacks:** Production deployments use Blue-Green deployments via ArgoCD. Automated canary analysis monitors error rates (HTTP 5xx) for 15 minutes. If error rates exceed 0.5%, traffic automatically rolls back to the previous stable release.